Deep Learning-Based Planetary Crater Detection

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Outline

Introduction and Motivation

- Planetary crater detection is important for:
  - planetary surface analysis,
  - terrain characterization,
  - geological age estimation,
  - autonomous navigation,
  - future robotic exploration missions.
- Manual crater annotation is:
  - time-consuming,
  - inconsistent,
  - difficult at large scale.
- Goal of this project:
  - Develop a deep learning-based crater detection pipeline.
  - Compare multiple modern object detection architectures.
  - Evaluate crater localization performance on planetary imagery.

Project Objectives

- Build a complete crater detection workflow.
- Verify YOLO-format crater ground truth.
- Train multiple deep learning object detectors.
- Compare detection performance quantitatively.
- Visualize crater localization predictions.
- Investigate strengths and limitations of different architectures.

Planetary Images $\rightarrow$ Deep Learning Detection $\rightarrow$ Crater Localization

Dataset Overview

Dataset Used:

Martian and Lunar Crater Detection Dataset

Split	Images	Label Files	Craters	Empty Labels
Train	98	98	681	9
Validation	26	26	202	2
Test	19	19	151	0

- Images contain planetary surface terrain.
- Labels provided in YOLO bounding-box format.
- Craters annotated using normalized coordinates.

Ground Truth Pipeline

YOLO Label Files

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Normalized Coordinates

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Pixel Coordinate Conversion

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Bounding Box Rendering

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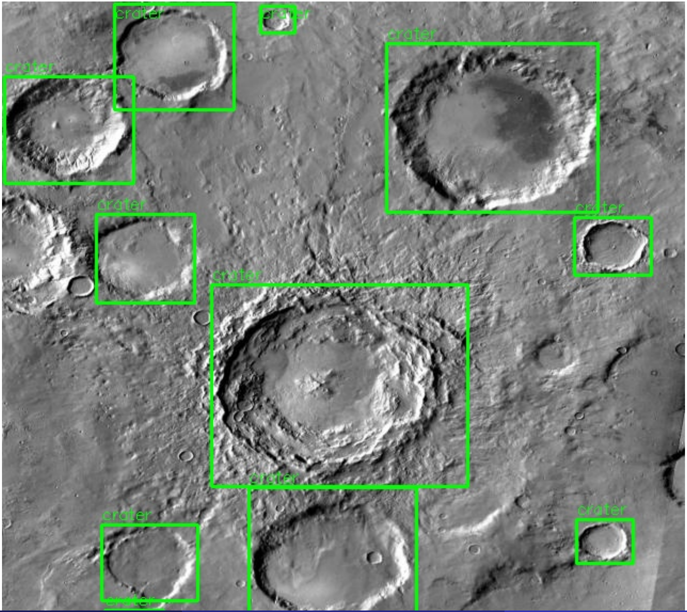
Ground Truth Visualization

Ground Truth Example



Related Work

- Classical crater detection methods:
  - edge detection,
  - Hough transforms,
  - template matching,
  - morphological operations.
- Modern crater detection methods:
  - CNN-based detectors,
  - segmentation networks,
  - YOLO architectures,
  - transformer-based vision systems.
- Recent research trends:
  - foundation models,
  - multimodal vision systems,
  - automated planetary terrain analysis.

Overall Detection Pipeline

Planetary Surface Images

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Ground Truth Verification

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YOLOv8 / Faster R-CNN / RetinaNet

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Model Training

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Prediction Generation

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Evaluation and Comparison

YOLOv8

- One-stage object detector.
- Simultaneous classification and localization.
- Lightweight and efficient architecture.
- Used pretrained `yolov8n.pt` weights.
- Trained for 50 epochs.

Input Image

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Feature Extraction

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Bounding Box Prediction

Faster R-CNN and RetinaNet

Faster R-CNN

- Two-stage detector.
- Uses Region Proposal Network.
- Strong localization refinement.
- ResNet50-FPN backbone.

RetinaNet

- One-stage detector.
- Uses focal loss.
- Designed for class imbalance.
- ResNet50-FPN backbone.

Transfer Learning with Pretrained COCO Weights

Training Configuration

Model	Epochs	Batch Size	Framework
YOLOv8	50	8	Ultralytics
Faster R-CNN	3	2	torchvision
RetinaNet	3	2	torchvision

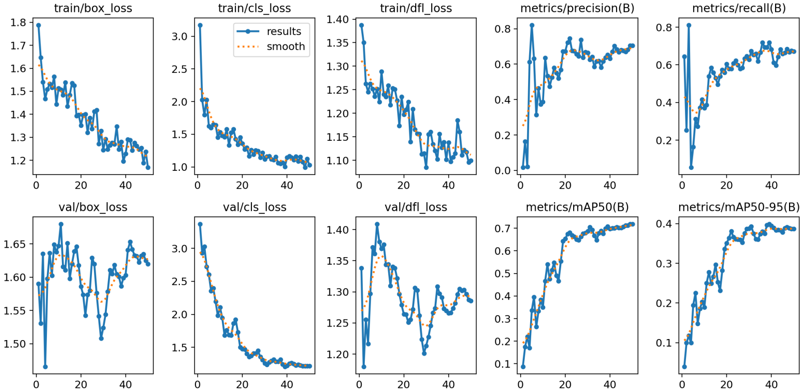
- Transfer learning used for all models.
- YOLO-format crater labels converted for torchvision detectors.
- Evaluation metrics:
  - Precision,
  - Recall,
  - mAP@0.5,
  - mAP@0.5:0.95.

Quantitative Results

Model	mAP@0.5	mAP@0.5:0.95	mAP@0.75
YOLOv8	0.700	0.366	0.374
Faster R-CNN	0.691	0.354	0.314
RetinaNet	0.002	0.001	0.000

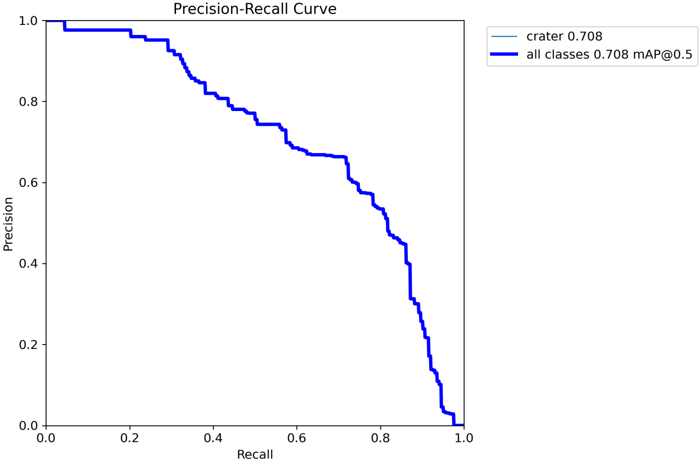
- YOLOv8 achieved strongest overall performance.
- Faster R-CNN achieved competitive localization.
- RetinaNet struggled under limited-data conditions.

Training Curves



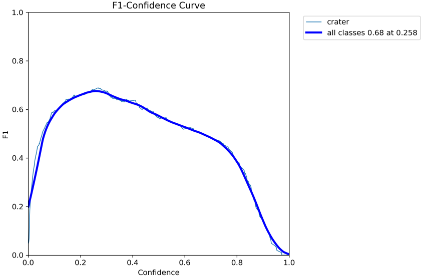
- Stable convergence observed during YOLOv8 training.
- Validation metrics improved progressively.

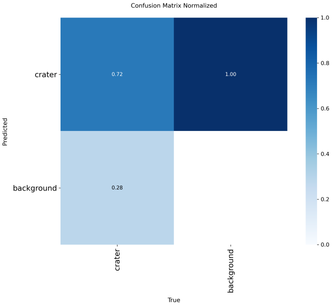
Precision-Recall Curve



- Strong precision-recall behavior.
- Detector maintained relatively low false-positive rate.

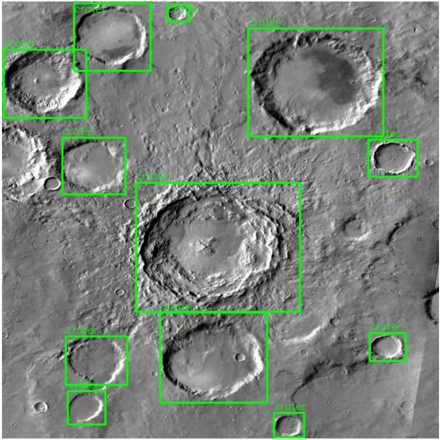
F1 Curve and Confusion Matrix



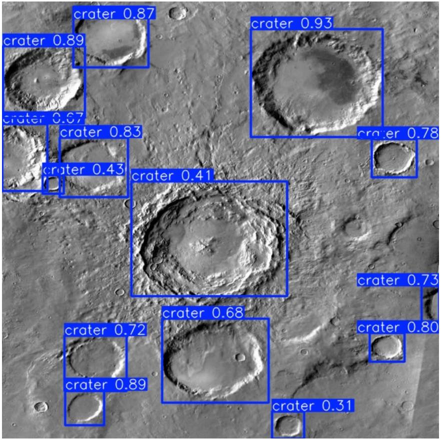


- Strong crater classification capability.
- Balanced precision and recall behavior.

Ground Truth

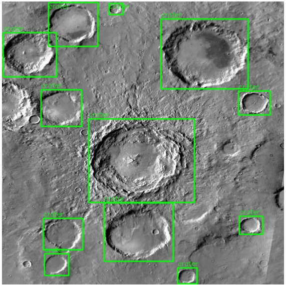


YOLOv8 Predictions

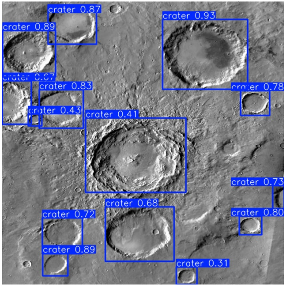


Detector Output Comparison

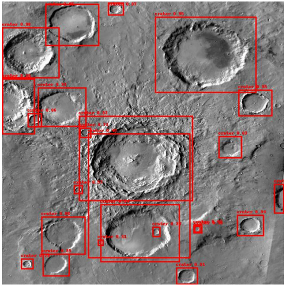
Ground Truth



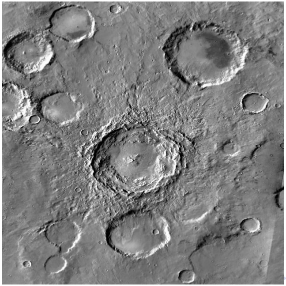
YOLOv8



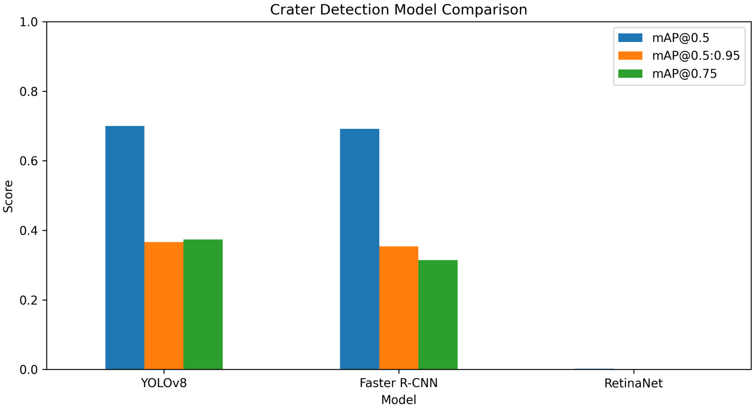
Faster R-CNN



RetinaNet



Final Model Comparison



- YOLOv8 achieved strongest crater localization.
- Faster R-CNN achieved similar performance.
- RetinaNet underperformed due to limited-data training.

Discussion

- YOLOv8 achieved strongest overall performance.
- One-stage architecture provided efficient detection.
- Faster R-CNN demonstrated strong localization refinement.
- RetinaNet struggled under limited-data conditions.

Major Crater Detection Challenges

- overlapping crater structures,
- degraded crater rims,
- low-contrast terrain,
- varying crater scales,
- illumination variability.

Future Work

- Larger crater datasets.
- Higher-resolution planetary imagery.
- Transformer-based object detectors.
- Foundation-model-based crater detection.
- Crater segmentation methods.
- Semi-supervised learning.
- Multi-class geological feature detection.

Future Direction: Foundation Models + Planetary Vision Systems

Conclusion

- Developed a complete crater detection pipeline.
- Verified and visualized YOLO-format ground truth.
- Trained and compared multiple deep learning detectors.
- YOLOv8 achieved best overall crater detection performance.
- Demonstrated feasibility of deep learning for planetary terrain analysis.

Thank You

References

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